MATHEUS LAJARIN ZEM

Backend Engineer · Java / Spring Boot

Piracicaba, Brazil (UTC−3) · Open to remote work, overlapping with US and European hours

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SUMMARY

Backend engineer with two years of production experience in Java and Spring Boot, currently responsible for more than thirty agribusiness systems integrated with the TOTVS PIMS ERP. I own deliveries end to end — from the requirement with the client to release and production support — across PostgreSQL, Oracle and SQL Server. Outside work I build and self-host full-stack products (Java, Go, React, Angular) on my own server, with CI/CD, automated tests and AI integrations, including MCP servers that let each user's AI assistant work with their data.

EXPERIENCE

LXTech — authorized TOTVS channel · *Backend Developer (Java), contractor* · Nov 2024 – present

- Develop and maintain 30+ Spring Boot systems in production for agribusiness companies: costs, cash flow, purchasing, suppliers, truck weighing and industrial production, integrated with the TOTVS PIMS ERP.
- Build REST APIs and new features for the product modules; investigate and fix production issues reported by consultants, and meet clients to align requirements.
- Model data and optimize queries on PostgreSQL, Oracle and SQL Server; integrate with SAP and other TOTVS systems; modernize legacy code.
- Ship through three environments with code review, automated releases and Dockerized services.

Java, Spring Boot, Angular, PostgreSQL, Oracle, SQL Server, Docker, Git, CI/CD

CNH Industrial · *Project Management Intern* · Jun 2022 – Jun 2024

- Managed parts across projects, from registration to supplier relations, and automated controls and data analysis in a multinational environment.

ElringKlinger · *Kaizen Intern* · Mar 2022 – May 2022

- Tracked Kaizen requests and automated reporting across departments.

SELECTED PROJECTS

Monevas — live at monevas.matheuszem.org (private code)
Personal finance and investment portfolio manager: bank statement import, category budgets, and tracking of Brazilian stocks, REITs and BDRs with AI-assisted analysis. Modular monolith with JWT auth and an MCP server so the user's AI assistant can query their own portfolio; a Python service feeds it public B3 and CVM market data (22M+ price snapshots). Java 21, Spring Boot 3, Spring Security, Flyway, PostgreSQL, Angular, Python, Docker, GitHub Actions.

Foreman — github.com/MatheusZem12/foreman (public)
Self-hosted issue tracker whose issues are carried out by coding agents (Claude Code, Codex): each run works in an isolated git worktree, with multi-repository merges that publish all or nothing, a rule engine for workflows, per-stage permissions and cost analytics. Go, PostgreSQL, React, TypeScript, Tailwind, Docker, end-to-end test suite.

Lexiquest — private, running on my homelab
Language learning platform: a React dashboard plus an MCP server so each user's own AI assistant (Claude, ChatGPT) can run conversations, grade skills from 0 to 100 and grow the vocabulary, while the server keeps the prompts, schemas and validation. Java 21, Spring Boot 3, PostgreSQL, React, TypeScript, Testcontainers, Playwright, GitHub Actions, Docker.

Monitorame — capstone project (private code)
Cold-chain monitoring: ESP32 units with temperature, humidity and GPS sensors report over MQTT, and the system tracks vehicles, shipments and drivers on a real-time map with street-level routing (OSRM), geofences and spreadsheet bulk import. Java, Spring Boot (15 Maven modules), Angular, PostgreSQL, MQTT, Docker, Playwright.

Homelab — the server that hosts the projects above
Per-project dev and staging environments: Jenkins for dev, GitHub Actions → GHCR → Watchtower for staging,

Cloudflare Tunnel with Zero Trust access, Tailscale and encrypted daily backups (restic + rclone).

SKILLS

Backend: Java 17/21, Spring Boot 2/3, Spring Security, Spring Data JPA/Hibernate, REST, microservices, RabbitMQ, Maven, Gradle · Go · Python (FastAPI, Django REST)

Data: PostgreSQL, Oracle, SQL Server, MySQL, MongoDB · data modeling, query tuning, Flyway

Frontend: Angular 17–22, React, TypeScript, Tailwind, PrimeNG, RxJS

Testing & DevOps: JUnit, Mockito, Testcontainers, Playwright · Docker, GitHub Actions, Jenkins, Nginx, Cloudflare Tunnel, Linux servers

AI tooling: MCP servers, LLM APIs (Anthropic, OpenAI, Google), Claude Code

EDUCATION

B.Eng. Computer Engineering — Escola de Engenharia de Piracicaba (EEP) · expected Dec 2026
Capstone: IoT architecture for environmental monitoring and real-time traceability in the cold chain (Monitorame).

LANGUAGES

Portuguese — native · English — advanced (professional working proficiency)